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Advancing the C-AICD: From Material Screening to System Level Design

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Abstract

This paper continues the development of the Chemical Autonomous Inflow Control Device (C-AICD), a novel concept that uses fluid-responsive, selectively dissolvable coatings to regulate flow in downhole environments. Building on earlier CFD and design studies by Asthana et al. (2023) and AlMusharraf et al. (2024), this work focuses on material screening, coating development, and system-level design to advance the technology toward laboratory testing and future deployment.

A structured material screening process was conducted to evaluate metal oxide powders (SiO_2, TiO_2, Fe_2O_3) for staged dissolution in oil–water mixtures. These materials were selected based on solubility trends and hydrophilic/hydrophobic behavior, confirmed through contact angle measurements and preliminary characterization (XRD, FTIR, DSC). A multi-layer core design was proposed, and fabrication trials were conducted using binder-assisted methods. A flow-compatible prototype fixture was designed to simulate annular horizontal flow, and a two-phase flow loop was developed for upcoming validation. Sensor placement, geometry, and flow conditioning were engineered to match field-like test conditions.

The selected metal oxides demonstrated distinct dissolution profiles aligned with targeted fluid compositions. Fe_2O_3 dissolved readily in oil-rich environments, TiO_2 responded at intermediate water cuts, and SiO_2 remained stable in water-dominant mixtures. This behavior supports their use as outer, middle, and inner shell materials in a staged shutoff design. While full multi-shell core construction and flow testing are ongoing, early characterization confirmed the structural and thermal suitability of the individual layers. CAD-based design and mechanical integration of the C-AICD prototype were completed using downhole relevant dimensions. A flow loop was developed with dedicated separation, multiphase pumping, and instrumentation systems. Together, these developments create a complete experimental platform to validate coating dissolution behavior and flow restriction response. This paper presents the foundational work that bridges concept to testable hardware.

This work introduces a chemical-based ICD platform that departs from conventional flow-property selectivity by responding to fluid chemistry. By focusing on staged material dissolution and passive mechanical response, the C-AICD offers a pathway to autonomous shutoff without moving parts. The paper establishes a material and design foundation critical to future validation and field translation of the concept.

Introduction

Managing water production remains a critical challenge in oil and gas wells. Excess water inflow reduces production efficiency and higher operation costs (Akbari et al. 2025; Zeng et al. 2015). While autonomous inflow control devices (AICDs) have been widely used to mitigate water production, their effectiveness depends heavily on physical property contrasts, typically viscosity or density differences between oil and water. In many field scenarios, these contrasts are not sufficient to achieve reliable selectivity.

To address this limitation, Asthana et al. (2023) introduced the concept of a Chemical Autonomous Inflow Control Device (C-AICD), which departs from conventional flow-based selectivity by responding to the chemical composition of the fluid. The core idea, which is described in detail in US 11,788,377B2, involved using materials that dissolve selectively in water, enabling a physical flow restriction to occur only when water is present at certain oil/water ratios (Al-Sulaiman et al. 2023). A follow-up study by AlMusharraf et al. (2024) further advanced this concept by presenting a CFD-informed design of a ported funnel and coated ball system and exploring potential layer configurations for chemical responsiveness.

This paper builds directly on those efforts. While field testing is still ahead, the current focus is on the design, material screening, and prototyping steps required to bring the C-AICD concept closer to practical implementation. The work includes a screening process to identify three metal oxides suitable for constructing multi-layer coatings that respond selectively to varying water/oil ratios. The metal oxides chosen were SiO_2, TiO_2, and Fe_2O_3. It also summarizes the coating development methods and early validation of the multi-shell core system, along with preliminary material characterization using XRD, FTIR, and DSC techniques. In parallel, a CAD-based C-AICD fixture was designed to accommodate horizontal annular flow, and a two-phase laboratory flow loop was developed to support future testing of the device. Although the flow loop experiments are still in progress, the developments presented in this work demonstrate meaningful progress in translating the C-AICD from a theoretical concept to a testable, engineered system.

Background and Related Work

Autonomous inflow control devices (AICDs) are widely deployed in oil and gas completions to manage water production by taking advantage of differences in fluid properties. These systems are typically passive and operate by restricting water inflow based on its lower viscosity or density compared to hydrocarbons (Garcia et al. 2025). Designs may include channels, nozzles, or helical flow paths that create pressure drops favoring oil while impeding water.

While effective in many applications, viscosity- and density-based AICDs have limitations, particularly in wells where oil and water properties overlap or vary over time. These conditions are common in horizontal sections and heterogeneous formations. More importantly, conventional AICDs offer fixed flow responses and cannot adjust to fluid composition changes during a well's life. These challenges have driven interest in alternative approaches that can offer selective, chemical-based responsiveness.

The C-AICD was proposed by Asthana et al. (2023) as a new class of passive inflow control technology. Unlike conventional devices, the C-AICD is designed to react to the presence of water through chemical dissolution. The system uses a coated spherical component housed in a funnel geometry. When exposed to water, the coating gradually dissolves, eventually causing the ball to shift and block flow, providing an autonomous shutoff mechanism without requiring moving parts or power.

In a subsequent study, AlMusharraf et al. (2024) expanded on this idea by introducing a component-level design and validating the concept through CFD simulations. The study examined flow dynamics through the funnel-ball system. This work positioned the C-AICD as a potentially field-deployable solution for water control.

The present study continues the development of the C-AICD by transitioning from modeling and design into materials engineering and experimental prototyping. It focuses on the screening and selection of metal oxides with tailored dissolution behavior in oil–water environments, the development of scalable methods for constructing multi-shell cores using these materials, and the verification of coating integrity and thermal stability through standard material characterization techniques. The work also includes the design of a custom test fixture compatible with multiphase flow loops for future lab-scale validation.

While this paper does not include CFD or field test results, it serves as an important step toward realizing the chemical shutoff concept in a physical system. The work complements earlier publications by focusing on material and design integration and lays the foundation for upcoming flow loop testing.

Material Selection for Multishell Core

The performance of the C-AICD depends heavily on the ability to construct layered coatings that respond predictably to changes in fluid composition. This section outlines the materials selection process that guided the development of a three-layer oxide system and summarizes early-stage efforts to validate the structural and thermal suitability of those materials. While full multi-shell core fabrication is ongoing, the findings presented here represent the foundation for coating integration and system-level testing described in later sections.

Powder Screening and Multi-layer System Design

A key requirement for the C-AICD system is the ability to build a multi-shell coating structure around a spherical core, where each shell dissolves selectively in response to fluid chemistry. To identify suitable materials, a screening study was conducted on a range of metal oxide powders, namely SiO_2, TiO_2, Fe_2O_3, ZnO, MgO, and MnO_2, as well as polyethylene glycol (PEG) as a reference compound.

The goal of this part of the study was to assess the dissolution behavior of these powders in a range of oil–water mixtures under elevated temperature conditions. The data would support the selection of materials capable of providing staged reactivity: ideally, an outer shell that responds quickly in oil-rich conditions, an intermediate shell for moderate water content, and an inner shell that dissolves slowly in water-dominant flow.

Dissolution Testing Method. The dissolution behavior of each material was evaluated using a series of static tests. The powders were exposed to mixtures of seawater and oil at different volume ratios (ranging from 100% water to 100% oil) and stirred at 1500 rpm at 80°C for up to 18 days. Each sample started with 1 mg of powder, with incremental additions made if dissolution was observed. The solids were then separated, dried, and weighed to determine the extent of dissolution. This approach allowed for a simple comparative analysis across different fluid environments.

The dissolution profiles of the tested powders revealed varied behaviors across different oil–water ratios, as illustrated in Figure 1. While Fe_2O_3 showed relatively higher dissolution in oil-rich conditions and TiO_2 peaked at intermediate ratios, the response of SiO_2 was less distinct. It exhibited low and steady dissolution across all conditions, including water-dominant mixtures. Although SiO_2 did not show a sharp increase in solubility with water cut, its chemical stability and known hydrophilicity support its role as the innermost layer, intended to maintain structural integrity until late-stage water breakthrough. Other materials such as ZnO, MgO, and MnO_2 exhibited less distinctive trends or lacked the solubility contrasts needed for staged release and therefore were not selected for further development.

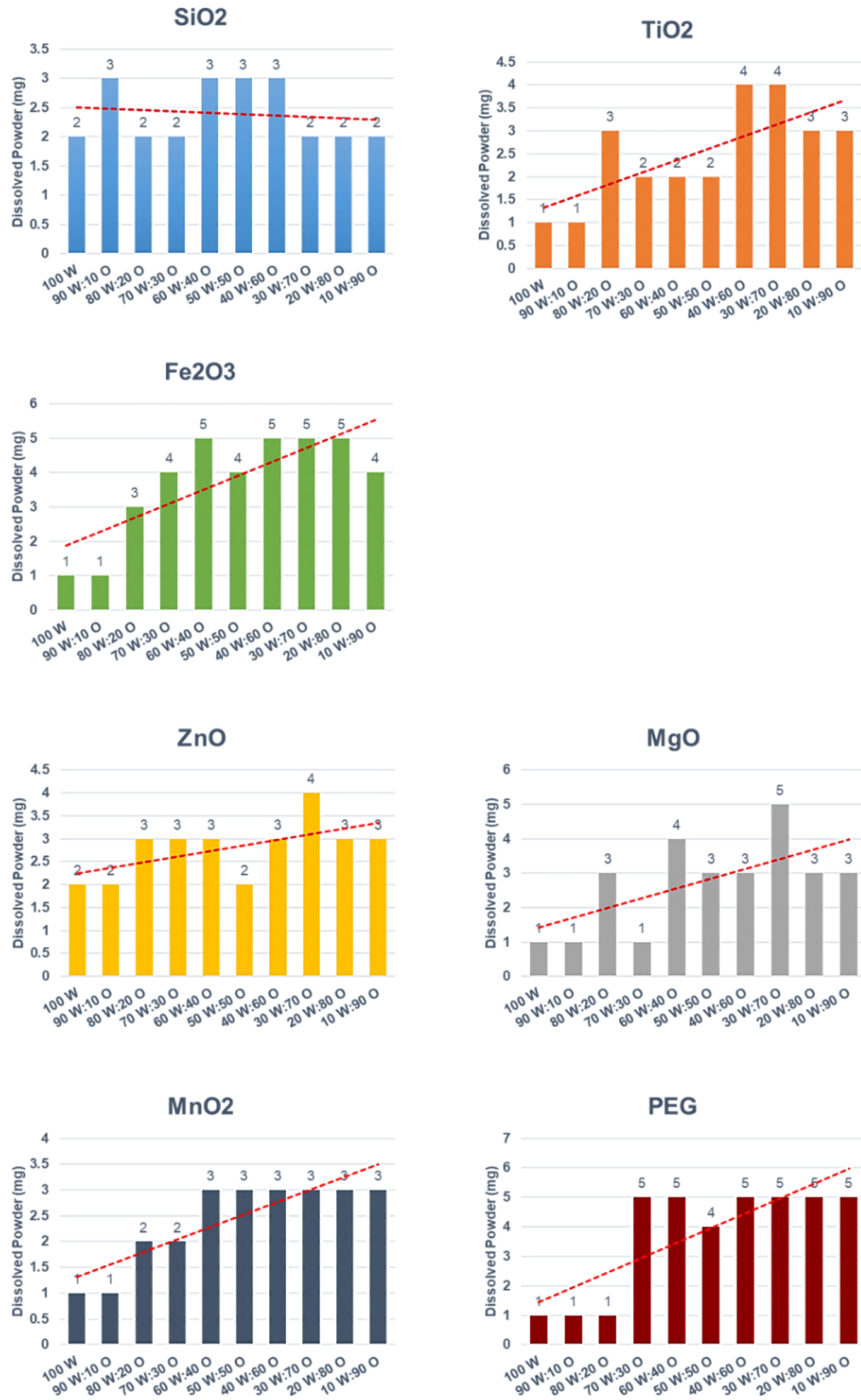

Figure 1—Dissolution results of various powders in different seawater/oil ratios mixed at 80°C for 18 days.

These trends were supported by contact angle measurements, which confirmed hydrophilic behavior for SiO_2 and TiO_2 (angles < 90°) and more hydrophobic tendencies for Fe_2O_3 (angle > 90°), as shown in Figure 2. This surface wettability aligns with the proposed dissolution order and helps reinforce the logic for layer sequencing.

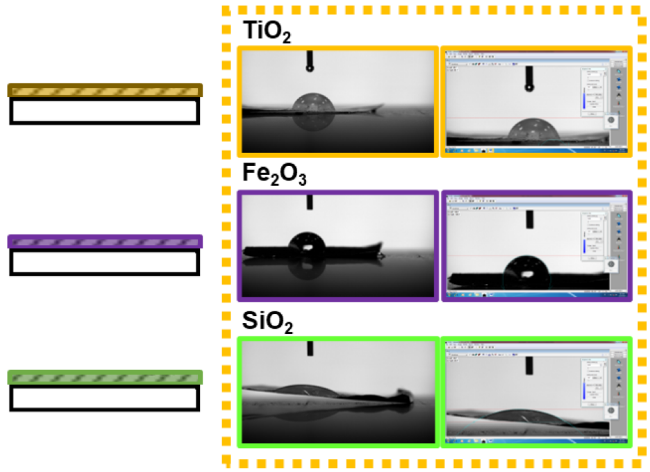

Figure 2—Contact angle measurements on the selected metal oxides, SiO_2, TiO_2, and Fe_2O_3.

Based on these findings, the working design for the C-AICD multi-layer system is as follows:

- Fe_2O_3 (outer layer): Dissolves first in oil-rich flow.
- TiO_2 (middle layer): Responds in intermediate water/oil conditions.
- SiO_2 (inner layer): Remains stable until high water cut is reached.

This sequence is designed to provide a gradual and directional shutoff behavior as the produced fluid evolves from oil-dominated to water-dominated over time. While this screening was based on powder solubility under static conditions, it offers a rational starting point for integration into the full core–shell fabrication process. Further development, including structural layer bonding and dissolution under dynamic flow, is addressed in later sections.

Overview of Material Characterization for Selected Oxides

To support the selection of SiO_2, TiO_2, and Fe_2O_3 as candidates for the C-AICD multi-layer system, a set of characterization studies were conducted to verify their structural and chemical properties. These included X-ray Diffraction (XRD), Fourier-Transform Infrared Spectroscopy (FTIR), and Differential Scanning Calorimetry (DSC), focusing on confirming phase identity, chemical bonding, and thermal stability. These analyses were performed during an earlier stage of the project, before the final material screening described in the previous section, as part of a broader effort to evaluate multiple oxide systems and binder interactions. The findings from this phase supported the use of these materials in the proposed shell arrangement and validated their compatibility with the coating process.

XRD confirmed that the selected powders exhibited expected phase purity and crystallinity. FTIR results showed characteristic bonding signatures, while DSC scans demonstrated that the materials remained thermally stable within the range expected for both lab-scale and downhole conditions. Given that this work preceded the focused dissolution study presented in this paper study, and that a larger dataset exists covering a wider range of formulations, only a summary is included here. Detailed results, including comparative spectra and thermal analysis, will be presented in a separate publication focused on coating development and process optimization. The key outcome of this phase was the confirmation that SiO_2, TiO_2, and Fe_2O_3 meet the structural and thermal requirements for inclusion in a chemically responsive multi-shell core system, supporting their use in the design and prototyping efforts described in the next sections. Figure 3 below shows the XRD, FTIR, and DSC analysis done on the selected metal oxides.

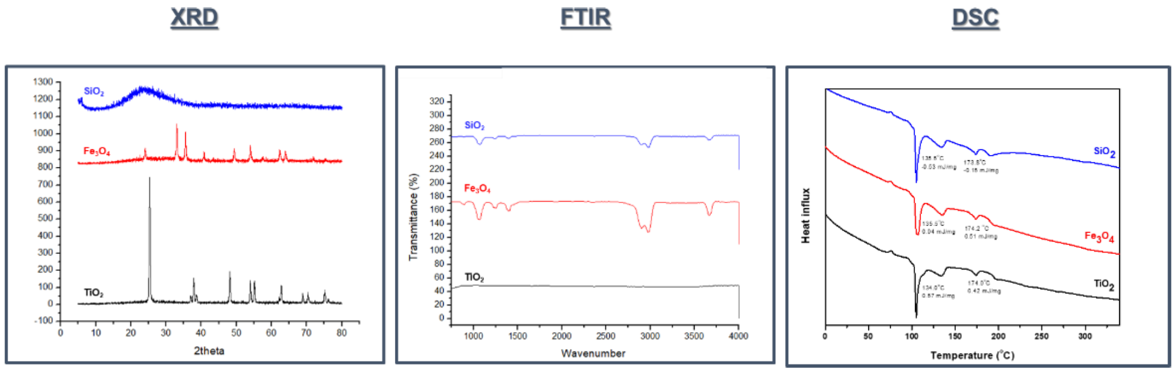

Figure 3—XRD, FTIR, and DSC for the selected metal oxides: SiO_2, TiO_2, and Fe_2O_3.

Chemical AICD System Design and Experimental Setup

This section describes the design of the C-AICD fixture and the custom-built flow loop developed to support future validation. Together, these systems provide the mechanical and experimental foundation for evaluating the chemical response and shutoff behavior of the device under multiphase flow conditions.

Mechanical Design and CAD Model of the C-AICD Prototype

A custom fixture was designed and fabricated to enable flow loop testing of the C-AICD under conditions that simulate realistic downhole flow. The fixture replicates the annular geometry commonly found in cased-hole or open-hole completions, using a concentric pipe arrangement that supports horizontal, two-phase flow (oil and water). The design also ensures compatibility with the coated core–funnel system that underpins the C-AICD concept.

The fixture consists of two main components: a modified inner pipe which serves as the base for flow and instrumentation, and a funnel which holds the chemically coated ball that reacts to fluid composition. CAD model drawings of these components are illustrated in Figures 4–6. The overall flow direction is horizontal, consistent with field completions and with prior CFD models (AlMusharraf et al. 2024). Fluid enters through the annular space between a larger outer pipe and a smaller inner pipe. Midway through the inner pipe, there is a local increase in wall thickness, followed by a sharp 90° reduction. This thicker middle section is specifically designed to house the funnel insert and allow precision machining of access ports. The funnel component, when inserted, directs flow over the coated ball. As the ball's multi-layer coatings dissolve in response to water or oil, its interaction with the funnel alters the effective flow area. This configuration enables controlled shutoff behavior based on fluid chemistry, not flowrate or density.

Threaded regions are integrated at both ends of the fixture. At one end, a threaded plug can be installed to block straight-through flow and force the fluid to pass through the annular section. At the other end, external threads allow secure connection to the broader flow loop setup using a retaining or coupling component. To manage sealing and prevent bypass, O-ring grooves are machined at multiple points, including above the funnel seat. These seals ensure that flow is directed through the funnel-ball interface, allowing for reliable observation of pressure drop and flow changes during testing.

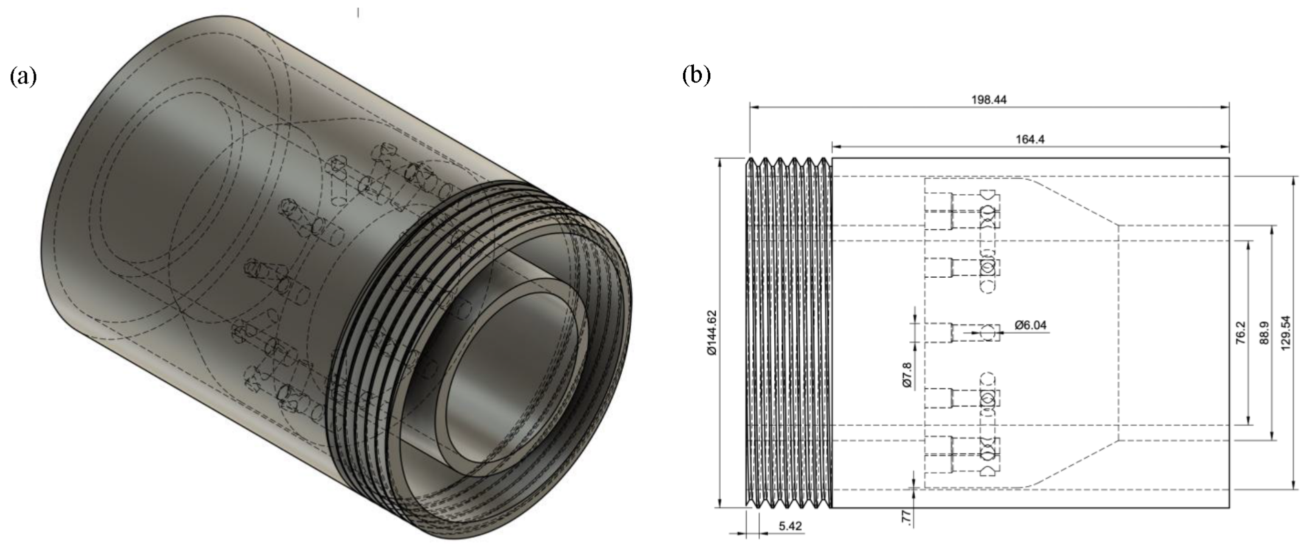

Figure 4—Design of the C-AICD fixture, (a) isometric view of the 3D CAD model, (b) detailed 2D drawing with dimensions.

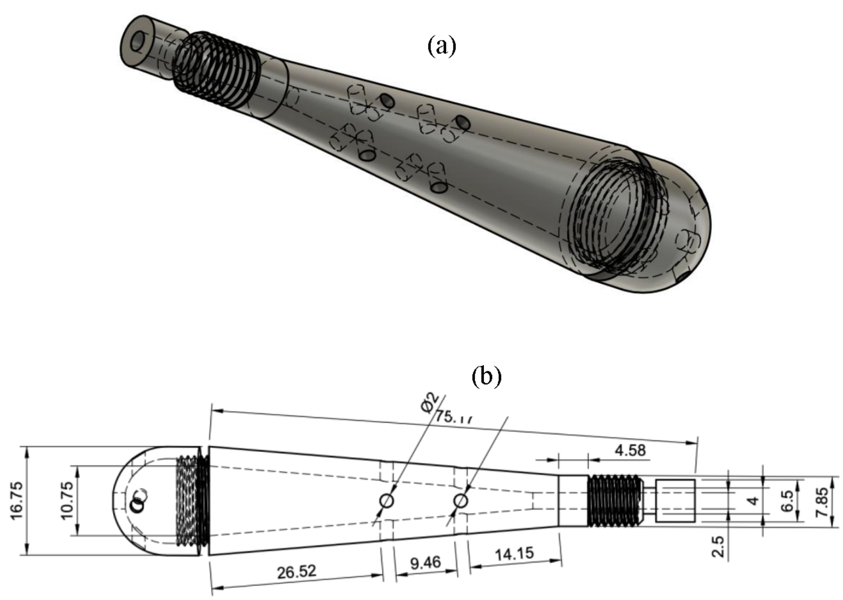

Figure 5—Design of the C-AICD funnel, (a) isometric view of the 3D CAD model, (b) detailed 2D drawing with dimensions.

(a) (b)

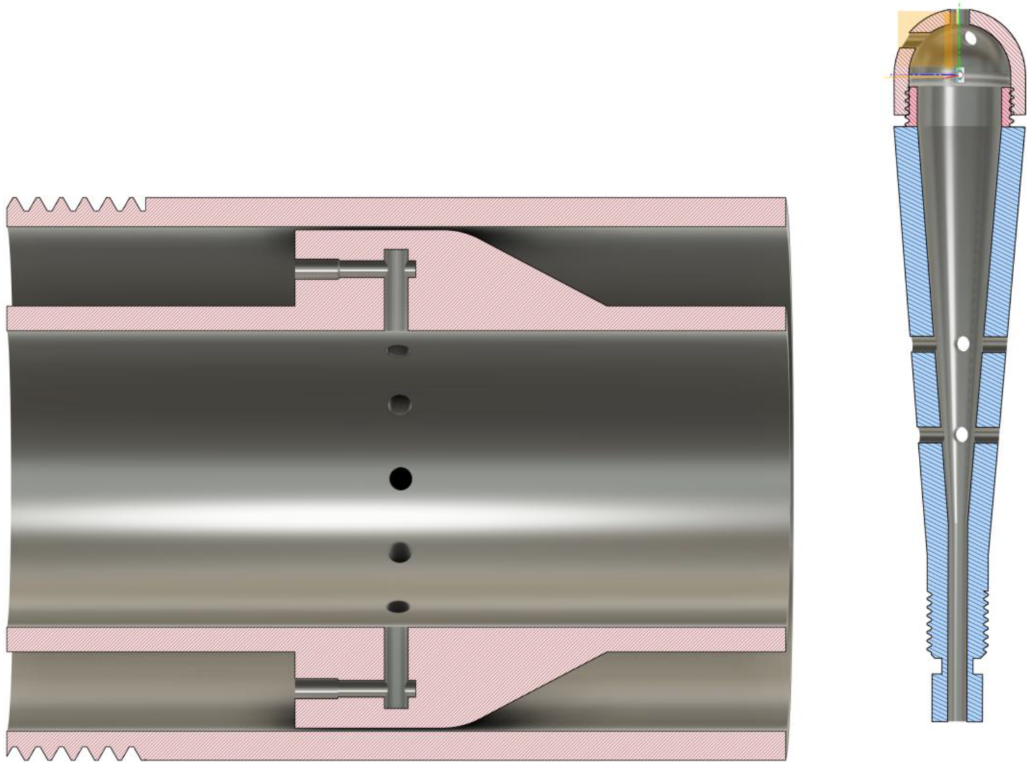

Figure 6—Cross section images of the C-AICD fixture (a) and funnel (b).

The fixture also includes allowances for adding temperature and pressure sensors immediately upstream and downstream of the C-AICD region, enabling real-time tracking of the device's response. The fixture components were designed for fabrication from stainless steel or carbon steel, both of which are common in oilfield equipment due to their strength, corrosion resistance, and compatibility with water/oil mixtures. The geometry and tolerances allow for straightforward machining using conventional CNC tools, and the fixture can be disassembled for inspection or cleaning between tests. The fixture is designed for use in moderate pressure and elevated temperature flow loop conditions, targeting a range of up to 300 psi and approximately 80–120 °C. These conditions are sufficient to evaluate the coating behavior without requiring full downhole-rated hardware.

Flow Loop Experimental Setup

A dedicated flow loop is being developed to evaluate the performance of the C-AICD prototype under controlled, multiphase flow conditions. The setup is designed to simulate oil–water flow in a horizontal annular configuration and will allow real-time monitoring of how the chemically responsive device behaves as fluid composition changes. Figure 7 illustrates the future flow loop setup.

The loop includes a large gravity separation tank, which acts as a reservoir for both oil and water phases. As fluid circulates through the system, it returns to this tank, where the two phases naturally separate. Two independent pumps are used to withdraw oil and water separately from the tank and inject them into the flow loop at controlled flowrates. This design allows flexible control over fluid ratios and simulates realistic phase behavior during production. Flow moves horizontally through 3-inch diameter piping, with crossover fittings used to connect the loop to the larger-diameter C-AICD fixture described in the previous section. The inner pipe of the fixture receives flow from the annular space and guides it through the funnel-ball assembly. The outlet section returns fluid to the top of the separation tank, minimizing turbulence and preserving clear oil–water separation.

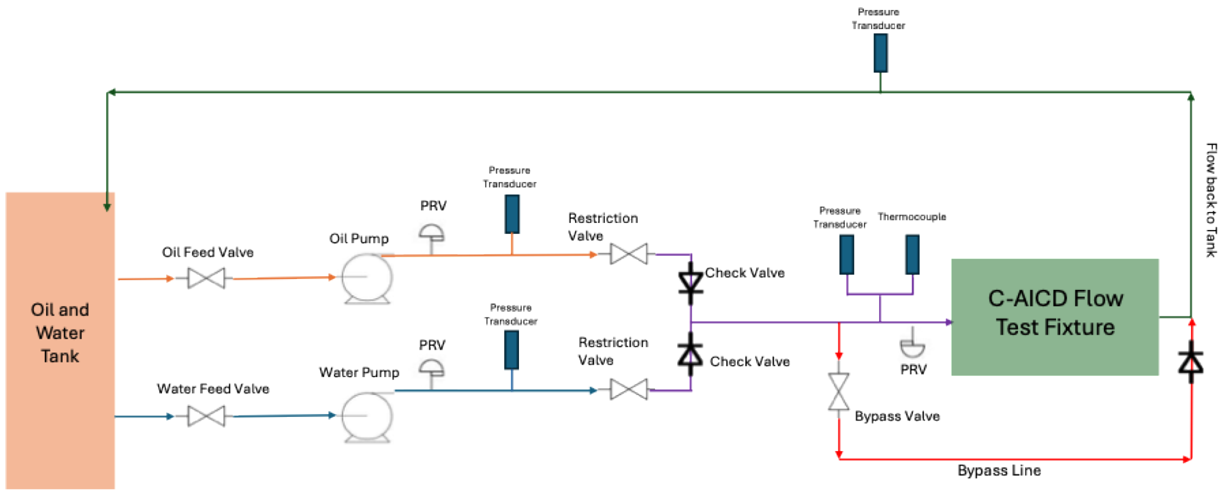

Figure 7—Flow loop set up for future testing of the C-AICD fixture.

To evaluate the flow response of the device, several sensors will be installed at key points in the loop. Pressure sensors will be positioned immediately upstream and downstream of the C-AICD fixture to measure the pressure drop across the system and identify flow restriction behavior. Temperature sensors will be placed before the fixture and within the fluid feed lines to ensure consistent delivery conditions for both oil and water phases. Additionally, a flowrate sensor will be installed upstream of the fixture to monitor overall flow and detect any changes that could indicate partial or complete blockage due to coating dissolution. Together, these sensors will provide real-time data on device activation and allow for correlation between flow behavior and the chemical response of the coated ball.

The flow loop is designed to operate under moderate pressure and elevated temperature conditions typical of lab-scale multiphase testing. The system will accommodate temperatures ranging from approximately 80 °C to 120 °C and pressures up to 300 psi. Flowrates will be adjustable to control residence time and simulate different flow regimes. The loop is capable of circulating oil-rich, water-rich, or mixed fluid systems, enabling the C-AICD to be tested under conditions that reflect early, mid, and late production stages in a reservoir. This flow loop setup serves as the final link in preparing the C-AICD for laboratory validation. It combines two-phase circulation, a representative mechanical fixture, and targeted instrumentation to enable future studies on C-AICD shutoff and coating dissolution timing. Once fully operational, this platform will allow quantitative evaluation of the chemical selectivity and performance repeatability of the C-AICD design.

Conclusion and Future Work

The development of the C-AICD has progressed significantly since its original introduction by Asthana et al. (2023) and the subsequent design and modeling work presented in AlMusharraf et al. (2024). This paper continues that trajectory by transitioning from conceptual design toward practical system integration, with a focus on material selection, coating development, and prototype fabrication.

A key outcome of this work is the establishment of a three-material coating system, Fe_2O_3, TiO_2, and SiO_2, designed to dissolve in response to specific fluid compositions. Through a structured screening study, each oxide was evaluated in oil–water mixtures to understand its dissolution trends. Based on this behavior, the materials were assigned to a sequence that supports staged shutoff: Fe_2O_3 as the outermost layer (oil-soluble), TiO_2 as the middle (responding to intermediate water cuts), and SiO_2 as the inner layer (chemically stable and hydrophilic). This sequence enables the device to react progressively as the produced fluid transitions from oil-rich to water-rich. While the screening was performed using static conditions, it provided a clear directional understanding of material performance, supporting future dynamic testing in the flow loop. This strategy emphasizes controlled selectivity based on chemical composition, rather than relying solely on viscosity or density differences, which is a key limitation of conventional AICDs. In addition to material selection, early-stage fabrication work confirmed the feasibility of constructing multi-layer coated spheres using binder-assisted milling techniques. Although a full multi-layer core was not completed in this phase, individual coatings were tested and characterized to validate adhesion and structural consistency. These findings reinforce the mechanical viability of the proposed design and establish a foundation for future core integration efforts.

On the mechanical side, the C-AICD fixture was fully designed and built using field-representative dimensions. The fixture mimics horizontal annular flow and includes dedicated features for installing the funnel-ball assembly, isolating flow, and integrating pressure and temperature sensors. The geometry closely follows earlier CFD-informed models, ensuring continuity with prior design logic even though no new simulations were performed in this study. Finally, the custom-designed flow loop system provides a platform for future validation. Its multiphase circulation, phase separation tank, and modular piping allow the introduction of different fluid compositions and flowrates. Once testing begins, it will enable direct observation of pressure response, coating dissolution timing, and shutoff performance.

While this paper does not include live experimental results, it demonstrates tangible progress in moving the C-AICD from a theoretical construct toward physical testing. The work lays the technical groundwork for a new class of chemically responsive inflow control devices capable of providing passive, staged water shutoff in dynamic reservoir environments.

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